Stat 5200: Analysis of Designed Experiments

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Overview

Objectives and overview

- This material is based Chapter 12 on the textbook "A First Course in Design and Analysis of Experiments" (Oehlert, 2000).
- Chapter outcomes include:
  - TODO

Nesting vs. Crossing

Nesting Versus Crossing

- Up to this point, our multi-factor designs have utilized crossed factors.
- **Crossed Factors:** Treatments are formed by all possible combinations of the levels of two or more factors. The same levels of Factor A are applied across all levels of Factor B.
- **Nested Factors:** Factor B is *nested* in Factor A if there is a completely different, unique set of levels of B for every level of A.
- **Example (Baby Carrots):** We want to test Vitamin A consistency across 3 brands of baby food. We buy 4 jars of each brand, and take 2 samples from each jar.
    - Jar 1 of Brand A has absolutely nothing to do with Jar 1 of Brand B.

Nesting Versus Crossing II

- Jars are **nested** within Brand. They do not cross.

The Nested Model

- Because Jar is nested in Brand, it makes no sense to estimate a "Main Effect of Jar 1" across all brands.
- If a main effect doesn't make sense, a crossed interaction (Brand $\times$ Jar) doesn't make sense either.
- We write nested models using parentheses in the subscripts:

$$y_{ijk} = \mu + \alpha_i + \beta_{j(i)} + \epsilon_{k(ij)}$$

- α_i: Fixed effect of Brand i.
- $\beta_{j(i)}$: Random effect of Jar j *nested within* Brand i.
- $\epsilon_{k(ij)}$: Random error of sample k nested in Jar j and Brand i.

Why do we use Nesting?

- We don't just "choose" to use a nested model. We use it because the experimental logistics or physical constraints dictate it.
- **Subsampling:** Taking smaller units from a larger selected bundle (Brands $\rightarrow$ Jars $\rightarrow$ Aliquots).
- **Logistical Constraints:** Suppose there are 3 specialized clinics in the country. We can't realistically send the same patient to all 3 clinics to "cross" patients with clinics. Patients are nested in clinics.

Why do we use Nesting? II

- **Biological Constraints:** Mating experiments. If we want to identify the sire (father), a female can only be mated to one male at a time. If females only live through one breeding, we must have completely different females for every male. Females are nested in males.

Nesting in R

The Nesting Coding Trap in R

- When recording nested data, you might number your jars $1, 2, 3, 4$ for Brand A, and $1, 2, 3, 4$ for Brand B.
- Warning: If you fit `(1 | Jar)`, `lme4` will assume Jar 1 in Brand A is the exact same entity as Jar 1 in Brand B (treating them as crossed).
- **The Fix:** Use the interaction operator `:` to explicitly define the unique combinations, or use the nesting shortcut `/`.

The Nesting Coding Trap in R II

```
library(lme4)
library(lmerTest)

# Assume 'carrots' data is loaded (Oehlert, Ch 12)
# Brand is Fixed. Jar is Random.

# INCORRECT: Treats Jar as crossed with Brand!
bad_mod <- lmer(VitA ~ Brand + (1 | Jar), data = carrots)

# CORRECT: Explicitly nests Jar within Brand
mod_carrots <- lmer(VitA ~ Brand + (1 | Brand:Jar), data = carro

# ALTERNATIVE CORRECT: shorthand for nesting
mod_carrots_alt <- lmer(VitA ~ Brand + (1 | Brand/Jar), data = c
```

Crossed and Nested Factors

Combining Crossed and Nested Factors

- Real-world designs often combine both structures.
- **Example (Gum Arabic - Oehlert 12.1):**
  - We want to test how emulsion shelf life is affected by acacia tree Variety (2 types) and Demineralization treatment (Yes/No).
  - We obtain 4 raw gum **samples** from each **Variety**. (Samples are nested in Variety).
  - Each sample is split into 2 subsamples. One gets Demineralized, the other does not. (Demineralization is **crossed** with Variety AND crossed with Sample).
- **The Model:**

$$y_{ijkl} = \mu + \alpha_i + \beta_{j(i)} + \gamma_k + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk(i)} + \epsilon_{l(ijk)}$$

Combining Crossed and Nested Factors II

- Notice: Fixed main effects (α, γ), Random nested effects $(\beta_{j(i)})$, Fixed crossed interaction $((\alpha\gamma))$, and Random mixed nested interaction $((\beta\gamma)_{jk(i)})$.

Fitting Crossed + Nested in lme4

- Translating the Gum Arabic design into `lme4` syntax requires specifying the fixed crossed terms and the random nested terms separately.

Fitting Crossed + Nested in lme4 II

```
# Assuming 'gum' data is loaded
# Variety (A) and Demin (C) are fixed and crossed.
# Sample (B) is random, nested in Variety.

mod_gum <- lmer(ShelfLife ~ Variety * Demin +
                  (1 | Variety:Sample) +
                  (1 | Variety:Sample:Demin),
                data = gum)

# (1 | Variety:Sample) accounts for the random variance of the s
# (1 | Variety:Sample:Demin) accounts for the random interaction

# how individual samples react differently to demineralization.
```

Hasse Diagrams

Visualizing Design: Hasse Diagrams

- As designs get complex, keeping track of what is nested inside what, and what crosses with what, becomes incredibly difficult.
- **Hasse Diagrams** (Lohr 1995) are graphical representations of a model showing the exact nesting/crossing and random/fixed structure.
- While historically used to manually derive Expected Mean Squares (EMS), today their primary value is helping you build your `lmer` formula without making structural mistakes.

Rules for Drawing a Hasse Diagram

1. Start at the top with a node **M** for the Grand Mean.
2. Add nodes below M for each factor that is not nested. Draw lines connecting them to M.
3. **Random factors go in parentheses:** e.g., (Jar)
4. For nested factors, draw them *below* the factor they nest in, connected by a line (e.g., Brand $\rightarrow$ (Jar)).
5. For crossed interactions, draw a node below the interacting main effects, with lines connecting up to both parents.
6. **Infection Rule:** Any term located below a random term in the diagram becomes random itself, and must be in parentheses!
7. End at the bottom with a node for **(E)** Error.

References and Acknowledgements

Oehlert GW (2000). *A First Course in Design and Analysis of Experiments*. W. H. Freeman. ISBN 0-7167-3510-5.

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